SEEIR: **S**emantic driven **E**nd-to-end **E**xploration and **I**nspection for optimizing 3D **R**econstruction Quality and Coverage

***Abstract*—We present SEEIR, a semantic-driven end-to-end framework that jointly explores unknown environments and actively inspects a specified target object to maximize visual coverage and fidelity of offline 3D reconstruction on an unmanned ground vehicle (UGV). The robot uses a navigation camera for safe exploration and an actuating camera pan-tilt (PT) inspection camera for high-resolution image capturing. This is achieved by optimizing a joint objective that reduces map uncertainty and increases surface-visibility coverage for efficient semantic target search. After target discovery, the agent transitions to active inspection and is rewarded using a bounded, diminishing-returns quality measure that encourages clear, close-up views with the PT camera. We train a multimodal policy in a photorealistic simulation using a three-channel egocentric map comprising occupancy, surface visibility, and visitation information. We evaluate SEEIR through extensive ablation studies demonstrating the impact of our multimodal fusion architecture and spatial memory channels. Furthermore, we qualitatively assess the policy's inspection performance by measuring mean surface coverage and analyzing reconstructed point clouds of non-primitive geometries.**

***Index Terms*—Reinforcement Learning(RL), Active Perception, 3D Reconstruction, 3D Gaussian Splatting, Robot Inspection and Exploration.**

I. Introduction

Autonomous inspection presents a unique challenge: robots must perform coverage planning and compute efficient paths that maximize coverage of the inspection target [1]. To address this, we propose SEEIR, a semantic-driven active perception algorithm that combines Deep Reinforcement Learning (DRL) with classical information-gain exploration for robotic inspection. By mounting an actuated Pan-Tilt (PT) camera on a UGV, SEEIR partially decouples viewpoint control from platform motion, allowing the agent to inspect the target from diverse orientations without repositioning its entire base.

The robot uses a forward-facing depth camera for navigation, an actuated Pan-Tilt inspection camera for high-resolution data acquisition and offline 3D reconstruction, and an IMU and odometry for state estimation. At the start of each episode, the robot is assigned a semantic target whose location is unknown. The agent must therefore explore the environment agnostically, identify the target via semantic observation, and safely navigate around obstacles and occlusions. Egocentric spatial representations provide reasoning about the observed environment and previously explored regions. After detecting the target, the agent uses active viewpoint control to maximize inspection coverage and acquire high-quality observations for offline reconstruction. We train SEEIR in randomized, cluttered warehouse environments and provide a GPU-accelerated RL benchmarking environment supporting 128 concurrent simulations per GPU.

A. Related Work

1) Learning for Autonomous Exploration and Navigation: Learning has emerged as a powerful paradigm for autonomous navigation, with Deep Reinforcement Learning (DRL) offering distinct advantages by bypassing the need for explicit, manually engineered heuristics and predefined trajectories [2]. These methods enable agents to learn direct mappings from high-dimensional sensor data, such as RGB-D images, to control actions or navigational waypoints through environmental interaction. By defining comprehensive reward functions, DRL agents can reason about diverse navigation tasks:

- Point-to-Point Navigation: Reward function is defined using success rates, distance-to-goal metrics, time, and collision penalties to ensure a safe and efficient path to a geometric coordinates [3], [4].
- Task-Agnostic Exploration: Utilizes reward functions based on information-theoretic objectives [5], [6], such as reducing map uncertainty via Shannon entropy or maximizing spatial coverage.
- Semantic Search and Inspection: Guided by target-driven rewards and contextual priors to locate specific objects.

DRL end-to-end learning capabilities offer significant flexibility; agents can either output high-level goals [7], [8] for semantic search or low-level control [4] commands for precise inspection. In particular, specialized "face reward" formulations [9] can be leveraged to maximize surface coverage and visual resolution in 3D reconstruction without requiring the complex path-planning paradigms used in classical methods.

2) Active Perception for 3D Reconstruction: High-fidelity active 3D reconstruction aims to produce accurate, highly detailed, and photorealistic digital representations of a scene or object, a capability critical for autonomous inspection systems [10]. Active perception for modern reconstruction algorithms, such as Neural Radiance Fields (NeRFs) [11] and 3D Gaussian Splatting (3DGS) [12], are computationally expensive as rendering and online neural network training are resource-intensive [13]. Furthermore, evaluating reconstruction quality typically requires computing the pixel-level rasterized loss between a rendered image and the ground truth [14], which is difficult to compute in real time during environmental interaction. Due to these challenges, it is currently impractical to jointly

evaluate 3D reconstruction during DRL policy training. This emphasizes the need for inspection reward functions that can effectively translate into high-quality offline 3D reconstruction from images collected during environment interaction.

3) Semantic Object Search and Inspection: Semantic reasoning can significantly improve exploration efficiency by biasing searches toward likely object locations. In structured environments, this is achieved by leveraging common-sense priors [2], [15]–[17] derived from the co-occurrence likelihood of objects' embeddings. However, many practical scenarios, such as disaster scenarios and unstructured environments [18], [19], lack reliable semantic context between the target and its surroundings. In these task-agnostic scenarios, the agent must perform a more exhaustive search by combining geometric and semantic information gains [20], [21], leveraging chance detection of the object via object classification models.

4) Active Perception via Actuated Cameras: We consider active camera control tasks in which camera actuation via Pan, Tilt (PT), and/or Zoom (PTZ) is important to task performance. Prior works [22], [23] have demonstrated that DRL can drastically benefit from active actuated camera control for robot navigation and object search. However, most applicable literature is limited to non-actuated cameras. Consequently, prior learning-based PTZ systems have remained narrow in scope, focusing primarily on active perception tasks such as 2D object tracking and surveillance. Recent works apply RL and machine vision to adaptive PTZ control to reduce reliance on hand-crafted heuristics [24], enhance visual perception for low-confidence object detection [25], [26], and actively track dynamic targets with variable speeds [27]. Despite these advancements, learning systems that jointly perform semantic object search and high-quality 3D reconstruction on an unmanned ground vehicle (UGV) have seen limited application.

Classical active-camera methods show that zoom capability can mitigate the spatial constraints faced by robots during inspection, enabling lower-cost trajectories while preserving high-quality observations [28]–[30]. In particular, PTZ cameras enable a UGV to remain in safer, lower-collision regions while continuously capturing detailed, high-resolution views necessary for inspection and reconstruction tasks. This work seeks to address gaps in learning-based control for robots equipped with PTZ cameras, particularly in semantic search and exploration.

B. SEEIR

Perhaps the most relevant prior work is [9], which employs DRL for semantics-aware inspection path planning. While their method uses an unmanned aerial vehicle, benefiting from substantially greater maneuverability than the UGV considered in our work, we extend these learning-based paradigms to ground vehicles. We argue that the use of an active, actuated camera remains uniquely beneficial in UGV settings because it can drastically reduce trajectory length and path cost while maintaining inspection quality. Furthermore, whereas [9] optimizes for maintaining a rigid, predefined inspection distance, our method imposes no such constraint. Instead, we draw inspiration from the submodular pixel-density objectives introduced in [31] for multi-robot filming, adapting their framework to a single-agent active inspection task. Because our inspection reward considers the distribution of pixels over the target object's surface, our approach accounts not only for distance but also for how the relative inclination of the surface affects the effective resolution.

The contributions of this work are as follows:

- **SEEIR**: An end-to-end DRL framework that jointly optimizes for task-agnostic exploration and high-fidelity semantic target inspection, capable of generalizing to targets with complex and variable geometries.
- **Inspection Reward**: A novel view reward designed to optimize data acquisition for high-fidelity offline 3D reconstruction that simultaneously maximizes viewpoint-normal alignment and surface pixel density while minimizing motion of the target object in the image frame.
- **Multimodal Architecture**: We develop an attention-based architecture that effectively fuses 2D visual, 3D spatial, and proprioceptive data, ensuring stable learning and robust navigation in cluttered environments.
- **Open-Source Environment**: We provide a high-performance, GPU-accelerated simulation environment built on Isaac Lab, featuring procedural warehouse generation and support for active Pan-Tilt-Zoom camera actuation to support further research.

II. Problem Formulation

Our objective is to autonomously navigate an unknown environment to locate a semantic target ξ, which is a 3D object with surface S. We represent the surface S as a discrete set of observable faces $F = \{f_1, f_2, ..., f_N\}$. The agent aims to maximize both the visual coverage of F and the cumulative inspection quality $Q(f_i) \in [0, 1]$ of each observed face f_i. We formulate the inspection task as a Partially Observable Markov Decision Process (POMDP).

Observation Space ($\mathcal{O}_t$): At each time step, the agent receives the multimodal observation $\mathcal{O}_t = [D_t, \Omega_t, \mathcal{M}_t, s_t]$. The visual observation $D_t = [D_t^{\text{nav}}, D_t^{\text{PT}}]$ represents the recent depth image from the forward-facing navigation and the pan–tilt (PT) inspection cameras, respectively. The binary mask Ω_t identifies pixels belonging to the semantic inspection target in the PT camera view. Consistent with the observations reported in [9], our preliminary experiments indicated that depth-only observations generalized more reliably to unseen target geometries than RGB or RGB-D inputs. Consequently, the final policy uses the two depth images and the semantic target mask without RGB observations. The local map $\mathcal{M}_t$ encodes the spatial information about the environment as detailed in Sec. III-A. The proprioceptive state $s_t = [p_t, q_t, v_t, \omega_t, a_{t-1}, \theta_t^{\text{pan}}, \theta_t^{\text{tilt}}]$, where p_t and q_t denote the robot's 3D position and quaternion orientation relative to the initial pose, v_t and ω_t denote its linear and angular velocities, a_{t-1} is the previous control action, and θ_t^{pan} and θ_t^{tilt} are the PT camera joint angles.

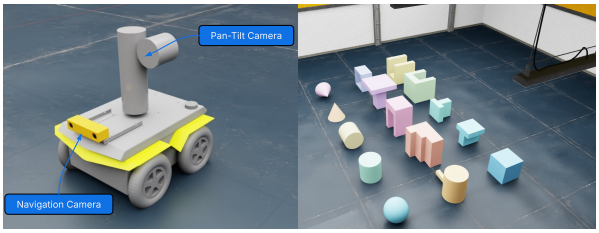


Fig. 1: Robot platform and inspection targets. (Left): The modified Clearpath Jackal UGV equipped with a forward-facing depth navigation camera and a custom actuated Pan-Tilt (PT) inspection camera. (Right): Primitive geometric objects used for training as inspection targets.

Action Space ($\mathcal{A}$): The continuous action vector $a_t \in \mathbb{R}^4$ simultaneously controls the differential-drive base and the PT camera joints. It is defined as $a_t = [a_t^{nav}, a_t^{PT}]$, where $a_t^{nav} = [v_t, \omega_t]$ commands the base's linear and angular velocities, and $a_t^{pt} = [\dot{\theta}_{pan}, \dot{\theta}_{tilt}]$ commands the PT camera's pan/tilt angular velocities.

III. Robot Inspection and Exploration

A. Environment Representation

To enable spatial awareness, the agent is provided with a multi-channel local, egocentric map of the environment's spatial features in its immediate surroundings.

1) Occupancy Channel $M_{\text{occ}} \in [-1, 1]$: To provide spatial information for collision-aware navigation, we utilize a forward-facing depth navigation camera mounted on the robot's base. Its depth observations are processed from this camera via ray casting to generate a 3D occupancy map. Each voxel v_i stores the occupancy log odds $\ell(v_i)$, which allows for additive updates as new measurements are taken. The log odds are clamped to $[-5, 5]$ and normalized before being provided to the policy.

2) Surface Visibility Channel $M_{vis} \in [0, 1]$: To drive semantic target search and prevent redundant inspection, we maintain a surface visibility map M_{vis} that tracks which surfaces the agent has successfully observed. We use depth data from the actuated pan-tilt (PT) depth camera to update the visibility state of surfaces. This provides a persistent memory of scanned surfaces, encouraging the robot to observe novel surfaces to locate the target ξ via semantic segmentation while ensuring thorough coverage during active inspection.

3) Visitation Map M_{visit}: To track the robot's trajectory, we maintain a 3D visitation map using the robot's odometry, inspired by spatial visit score representations in DRL exploration methods [5], [9]. This map serves as a curiosity bonus, encouraging the agent to traverse unexplored regions of the environment that might otherwise be missed by the navigation camera. Furthermore, during the active inspection phase, this representation incentivizes the robot to circumnavigate the target object. The map is updated by incrementing a voxel's visit count at each time step the robot occupies it.

B. Exploration and Semantic Object Search

The robot's objective is divided into three distinct components: exploring and mapping the unknown environment, locating the semantic target ξ, and performing high-quality visual inspection of its surfaces. These components are as follows:

1) Robot Exploration and Entropy Reduction: We employ an information-theoretic objective to reduce the robot's map entropy and uncertainty, thereby encouraging exploration and search for the inspection target ξ. This encourages the development of generalizable exploration policies and robust spatial representations.

Using the Shannon entropy definition, we define the entropy of a voxel v_i as the binary entropy of its occupancy probability p_i:

$$\mathcal{H}(v_i) = -\left[p_i \log(p_i) + (1 - p_i) \log(1 - p_i)\right] \tag{1}$$

Consequently, assuming independence across the voxel grid, the total entropy of the map M is the summation of the individual voxel entropies:

$$\mathcal{H}(M) = \sum_i \mathcal{H}(v_i) \tag{2}$$

2) Surface Visibility Coverage: While the entropy reduction objective described in Section III-B1 drives exploration via the navigation camera, it does not inherently encourage the PT camera to observe the inspection target. To bridge this gap, we introduce an objective that maximizes the PT camera's incremental surface-visibility coverage. To encourage comprehensive spatial coverage and penalize redundant views, we apply a saturation function to the observed surfaces. At time t, the current visibility score $\mathcal{S}(\mathcal{V}_t)$ is computed as a weighted sum over the visibility map cells v_i:

$$\mathcal{S}(\mathcal{V}_t) = \sum_{i=1}^{N} -e^{-k_v v_{i,t}} \tag{3}$$

where N represents the total number of tracked surface cells, $v_{i,t}$ denotes the cumulative visibility count of cell i up to time t, and $k_v > 0$ is a decay factor controlling saturation. This formulation naturally imposes diminishing returns for repeatedly observing the same surface, thereby incentivizing the agent to seek out and cover previously unobserved regions.

C. Inspection Quality Measure

Once the inspection target ξ comes into view, the agent seeks to maximize the quality of the observation of the target surface. For every observable mesh face f, we define a bounded quality value based on its projected pixel area, viewing angle, and image motion.

Recent learning-based inspection methods often impose geometric constraints, such as a preferred camera-to-target distance or a valid image region, to obtain high-resolution observations [9]. Other approaches approximate pixel density analytically as a function of distance [31]. In contrast, SEEIR

directly evaluates the projected appearance of each visible face, without requiring a hand-designed standoff distance. Let $\mathcal{P}_{f,t}$ denote the set of inspection-camera pixels observing face f at time t. For each pixel $p \in \mathcal{P}_{f,t}$, $\hat{\mathbf{n}}_f$ denotes the surface normal and $\hat{\mathbf{v}}_p$ denotes the viewing direction from the surface toward the camera. Their non-negative dot product favors observations made at a favorable incidence angle. Furthermore, to favor stable observations and penalize rapid image motion, we define a per-pixel optical-flow weight:

$$W_{\text{flow}}(p) = \exp\left[-\frac{1}{2}\left(\frac{\max\left(0, \|\mathbf{u}_p\|_2 - u_{\text{stable}}\right)}{\sigma_{\text{flow}}}\right)^2\right], \quad (4)$$

where $\mathbf{u}_p$ is the optical-flow vector at pixel p, u_{stable} is the motion threshold below which no penalty is applied, and σ_{flow} controls the decay rate. The angle and motion-weighted pixel count for face f at time t is given as

$$A_{f,t} = \sum_{p \in \mathcal{P}_{f,t}} \max\left(0, \hat{\mathbf{n}}_f^\top \hat{\mathbf{v}}_p\right) W_{\text{flow}}(p). \quad (5)$$

This formulation rewards viewpoints where a face occupies more pixels per frame (pixel-density) and is observed at a favorable incidence angle. Closer, unobstructed observations generally produce a larger weighted pixel count without requiring an explicit preferred distance. Moreover, PT actuation allows the agent to improve the viewing direction and projected coverage without relying exclusively on the motion of the UGV base.

We convert the weighted pixel count into a bounded per-frame quality score using a diminishing-returns function:

$$q_{f,t} = 1 - \exp\left(-\frac{A_{f,t}}{k}\right), \qquad k = 30. \quad (6)$$

Rather than accumulating repeated observations of the same face, the quality function strictly only updates its best quality:

$$Q_f(t) = \max\left(Q_f(t-1), q_{f,t}\right). \quad (7)$$

Consequently, repeatedly observing a face at the same or lower quality produces no additional inspection reward. The summed improvement across visible faces is normalized by the target's configured inspectable-face count when computing the inspection reward.

D. Joint Exploration Objective for Object Search

To explore the unknown environment and locate our inspection target ξ, we aim to learn a policy π that maximizes the joint objective of increasing visibility coverage while reducing map uncertainty. We define the total objective J as:

$$J(\pi) = \mathcal{S}(\mathcal{V}_t) - \lambda \mathcal{H}(M) \quad (8)$$

where $\lambda > 0$ is a weighting coefficient that balances the trade-off between surface coverage maximization and entropy reduction. Our ultimate goal is to find the optimal policy $\pi^* = \arg\max_\pi J(\pi)$.

IV. Methodology

A. Learning Joint Policy for Inspection and Navigation

1) Reward Design: To learn a policy that effectively balances exploration, safe navigation, and high-fidelity inspection, we formulate a dense reward function. The overall reward $R(s_t, a_t)$ obtained at each time step t is a weighted sum of individual objectives

$$R(s_t, a_t) = \gamma_E R_{\text{exploration}} + \gamma_V R_{\text{visibility}} + \gamma_T R_{\text{visitation}} + \gamma_I R_{\text{inspect}} + R_{\text{success}} - R_{\text{penalty}}, \quad (9)$$

where γ_E, γ_V, γ_T, and γ_I are scaling coefficients for the corresponding reward terms. R_{success} is a completion bonus awarded when the current curriculum coverage threshold is reached.

2) Exploration Reward $R_{exploration}$: To incentivize the agent to reduce its uncertainty about the environment geometry, we provide an information-theoretic reward based on the global occupancy map $\mathcal{M}$:

$$R_{\text{exploration}} = \left[\mathcal{H}\left(\mathcal{M}_{t-1}^{\text{occ}}\right) - \mathcal{H}(\mathcal{M}_t^{\text{occ}})\right]_+, \quad (10)$$

where $[x]_+ = \max(0, x)$. Thus, the exploration reward is the non-negative reduction in the entropy of the global occupancy map between consecutive updates.

3) Visibility Reward $R_{\text{visibility}}$: To encourage the PT camera to observe previously unseen surfaces, we define the visibility reward as the non-negative increase in the saturated visibility-map score:

$$R_{\text{visibility}} = \left[\mathcal{S}\left(\mathcal{M}_t^{\text{vis}}\right) - \mathcal{S}\left(\mathcal{M}_{t-1}^{\text{vis}}\right)\right]_+, \quad (11)$$

where $[x]_+ = \max(0, x)$. Because $\mathcal{S}$ saturates with repeated observations, the reward favors newly observed surfaces while providing diminishing returns for redundant viewpoints.

4) Visitation Reward $R_{\text{visitation}}$: To encourage the robot to traverse unexplored spatial regions and circumnavigate inspection targets, we incorporate a visitation-based curiosity bonus. Let i_t denote the voxel occupied by the robot at time t, and let $N_{i_t,t}$ denote the number of previous visits to that voxel before the current map update. The visitation reward is

$$R_{\text{visitation}} = \exp\left(-\beta N_{i_t,t}\right), \quad (12)$$

where $\beta > 0$ controls the rate at which the reward decreases with repeated visits.

5) Inspection Reward $R_{inspect}$: Using the per-face quality defined in Sec. III-C, we define the normalized target-quality score as

$$\bar{Q}_\xi(t) = \frac{1}{N_\xi} \sum_{f \in \mathcal{F}_\xi} Q_f(t), \quad (13)$$

where $\mathcal{F}_\xi$ is the set of tracked target faces and N_ξ is the target's configured inspectable-face count. The inspection reward is the marginal increase in this score:

$$R_{\text{inspect}}(t) = \left[\bar{Q}_\xi(t) - \bar{Q}_\xi(t-1)\right], \quad (14)$$

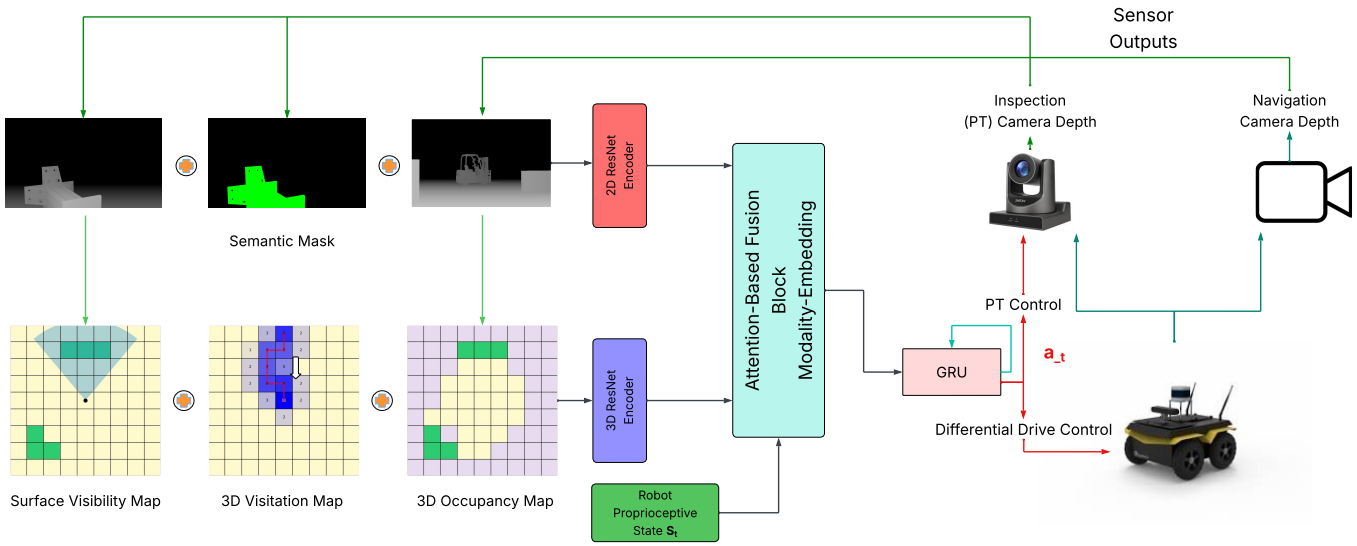


Fig. 2: Overview of the SEEIR architecture. 2D visual and 3D spatial features are extracted using ResNet encoders and fused with the robot's proprioceptive state s_t via a Transformer-based cross-sensor attention block. A GRU layer processes the resulting embeddings to incorporate temporal context, outputting the 4 continuous control actions for the robot base angular and linear velocity and PT-camera pan and tilt velocity)

6) Action and Safety Penalties $R_{penalty}$: To ensure safe navigation and discourage erratic camera movements, we apply a composite penalty function:

$$R_{\text{penalty}} = \lambda_{\text{base}}\|\Delta\mathbf{u}_{t,\text{base}}\|_2^2 + \lambda_{\text{PT}}\|\Delta\mathbf{u}_{t,\text{PT}}\|_2^2 + \lambda_{\text{col}}C_t + \lambda_{\text{time}} \tag{15}$$

To discourage jitter behavior, we penalize the squared L_2 norm of the temporal differences in consecutive control actions for both the mobile base ($\Delta\mathbf{u}_{t,\text{base}}$) and the PT camera ($\Delta\mathbf{u}_{t,\text{PT}}$). Furthermore, to ensure safe navigation, the environment applies a multi collision penalty: a map-based proxy penalty is applied if the agent approaches within a bounding volume of an occupied voxel in M_{occ} reflecting its physical dimensions, while a primary collision penalty λ_{col} is triggered if the robot's contact forces exceed a predefined threshold. Additionally, λ_{time} is a constant time-step penalty designed to encourage efficient task completion."

B. Neural Network Architecture and inputs

Our agent processes multi-modal inputs designed for spatial and temporal reasoning (see Fig. 2).

1) 2D Visual Inputs: The navigation-camera and inspection camera depth images, and the binary target semantic mask are stacked to form a three-channel 128×128 observation. This is processed by a 2D ResNet feature encoder.

2) 3D Spatial Mapping:: The policy receives an egocentric local map $M_t \in \mathbb{R}^{29\times29\times15\times3}$ centered on the robot. The map has a voxel resolution of $0.1\,\text{m}$ and therefore covers a $2.9 \times 2.9 \times 1.5\,\text{m}$ volume. Its three channels represent occupancy, surface visibility, and visitation. The resulting tensor is processed by a 3D ResNet feature encoder.

C. Modality Embedding and Fusion

To effectively fuse the distinct observation spaces, we employ a multimodal Transformer architecture [32], [33]. As illustrated in the system architecture Fig. 2 and detailed in the fusion module Fig. 3, high-dimensional features from the camera and map encoders, along with the low-dimensional proprioceptive robot state, are independently projected via linear layers into a uniform embedding space of dimension $d_{model} = 512$. These projected features are treated as a sequence of modality tokens:

- **Camera Token:** Encodes the features extracted from the navigation, inspection depth images, and target semantic mask.
- **Map Token:** Encodes the 3D spatial occupancy, visibility, and visitation features from the egocentric map.
- **Proprioceptive Token:** Encodes the robot's instantaneous state, including pose, velocity, and PT Camera joint positions, as well as control action from the previous timestep.

To stabilize training, we apply Layer Normalization to the sequence token before adding learnable modality embeddings, which allows the model to distinguish between sensor sources. The resulting token sequence is then processed by a two-layer Transformer encoder with four self-attention heads. This architecture allows SEEIR to capture cross-modality dependencies, enabling each modality to conditionally attend to features from the others. Finally, we apply average pooling across the output token sequence to produce a unified feature representation for temporal processing.

D. Temporal Processing

The unified representation produced by the fusion module in IV-C is passed to a single-layer Gated Recurrent Unit (GRU)

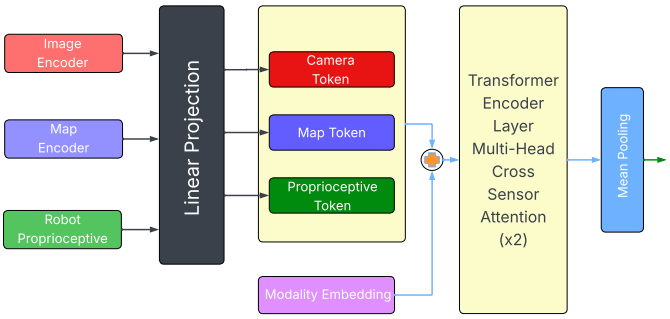


Fig. 3: Attention module used for multimodal fusion

with 512 hidden units. We use a sequence length of 32 time steps to model the temporal context. The GRU output is then fed into separate MLP heads to generate the policy distribution and value estimate.

V. Implementation

We train our policy using a recurrent Proximal Policy Optimization (PPO) algorithm, specifically leveraging the PPO-GRU implementation provided by the SKRL reinforcement learning library [34]. To encode high-dimensional multimodal observations, including local 3D spatial maps and 2D visual/semantic streams from both cameras, we use 3D and 2D ResNet architectures inspired by IMPALA [35]. As detailed in Section IV-C, these encoded features are passed through a Transformer-based attention module before temporal processing.

Furthermore, to facilitate stable learning in highly cluttered environments, we employ a curriculum learning strategy during training. Specifically, we progressively scale the number of active obstacles and increase the required surface coverage thresholds as the agent's success rate improves.

VI. Training Environment and Simulation

Talk about how the simulation is genertaes somewhere in this paraggrapsh perhaps we introduce additional constraints to the scene using basic geometric primitives (e.g., cones, spheres) with randomized positions and scales. Using a 2D grid, we procedurally generate the scene by uniformly sampling positions across the warehouse space for obstacles, inspection targets, and the agent's starting pose. To encourage generalization, the specific category, shape, and color of the obstacles are also randomized across episodes. Training is conducted on 2X NVIDIA RTX PRO 6000 Blackwell Workstation using Isaac Lab Simulation, leveraging the open-room Warehouse environment from the Isaac Lab asset suite.

We deploy a modified Clearpath Jackal UGV equipped with a custom actuated PT camera. The policy produces four normalized continuous actions: chassis linear velocity, chassis angular velocity, pan velocity, and tilt velocity. These commands jointly control robot motion and inspection viewpoint.

To achieve highly parallelized sensor simulation, the navigation and PTZ cameras utilize Isaac Lab's tiled rendering API. Additionally, to accurately track surface coverage, we employ a ray-casting camera to directly extract per-face hit information from the discretized mesh. Using GPU-accelerated ray casting and depth observations, we maintain three global representations for each episode: (i) an occupancy map, (ii) a surface-visibility map, and (iii) a visitation map. The global map bounds are set to $X \in [-10.5, 9.5]\,\mathrm{m}$, $Y \in [12.5, 18.0]\,\mathrm{m}$, and $Z \in [0.0, 2.5]\,\mathrm{m}$ (constrained by the ground vehicle's operational volume), with a voxel resolution of $0.25\,\mathrm{m}$. As established in our methodology, we extract an agent-centric local map from these global representations to serve as the policy input, significantly improving generalizability and mitigating the effects of global odometry drift.

The physics simulation operates at $129\,\mathrm{Hz}$. Actions are applied every 12 physics steps, resulting in a policy and map-update rate of $10.75\,\mathrm{Hz}$. The navigation camera operates at approximately $10.75\,\mathrm{Hz}$, while the inspection camera and face ray caster operate at $4.17\,\mathrm{Hz}$. During training, a performance-based curriculum progressively increases the environmental clutter and required inspection coverage. Each evaluation episode is limited to 1200 policy steps, corresponding to approximately $111.6\,\mathrm{s}$ of simulated time.

We apply Gaussian perturbations to position, orientation, and velocity observations. Depth observations receive depth-proportional Gaussian noise and random pixel dropout, while the semantic mask is corrupted using false-positive and false-negative pixel flips.

The Jackal is constrained to a maximum linear velocity of $2\,\mathrm{m/s}$ and an angular velocity of $4\,\mathrm{rad/s}$. Using GPU-accelerated ray tracing and depth from the navigation and inspection cameras, we maintain (i) a global occupancy map and (ii) a surface-visibility map for each episode. The global map bounds are $X \in [-10.5, 9.5]\,\mathrm{m}$, $Y \in [-12.5, 18.0]\,\mathrm{m}$, and $Z \in [0.0, 2.5]\,\mathrm{m}$. At a voxel resolution of $0.1\,\mathrm{m}$, the global maps have dimensions $200 \times 305 \times 25$. The simulation environment additionally supports discrete optical zoom through a configurable set of focal lengths. However, zoom is excluded from all experiments reported in this work because independently updating camera intrinsics introduces substantial overhead in massively parallel tiled rendering. Accordingly, the evaluated policy uses a fixed focal length and controls only chassis velocity, pan, and tilt.

VII. Experimentation and Ablation Studies

A. Experimental Setup

We evaluate SEEIR in the Isaac Lab Warehouse environment described in Section VI. We use a validation set of objects not present in the training mesh. We aimed for geometric diversity and complexity. We randomized the robot's and inspection target's positions and orientations. Similar to training, we also randomize the positions and types of the obstacles in the environment to serve as physical constraints and occlusion to the inspection target Because our primary objective is to maximize the visual information gathered by the PT camera, we quantify the agent's inspection performance by measuring the Mean Surface Coverage, defined as the fraction of the target's empirically calibrated inspectable mesh faces observed during an episode.

TABLE I: Ablation results averaged across five targets and four seeds. Each variant differs from baseline SEEIR only in the named component. Bold and underline indicate the best and second-best results.

Variant	Coverage ↑ (%)	Quality ↑ $[0, 1]$	Crash ↓ (%)
SEEIR	**90.07**	0.50	<u>3.63</u>
Map: OCC	<u>88.28</u>	<u>0.53</u>	4.22
Map: OCC + Visibility	77.08	**0.54**	20.47
Fusion: no MHA	85.82	0.51	10.39
Temporal: no GRU	86.66	0.50	37.93
Camera: fixed	60.87	0.52	**2.71**

B. Ablation Studies

To evaluate the contribution of individual components within our proposed architecture, we perform a series of ablation studies. For all ablation experiments, we train the respective models for $45e^6$ steps to assess sample efficiency and evaluate them over 128 episodes with 64 concurrent environments and 4 different seeds. We report the mean surface coverage achieved by the agent across 6 held-out inspection targets not in the training dataset.

1) Modality Fusion Ablation:MHA: We investigate whether explicitly modeling interactions among the camera, the map, and the robot-proprioceptive state improves policy performance. In preliminary trials, direct concatenation followed by an MLP exhibited unstable learning and consistently underperformed the token-based variants. We therefore adopted a token-wise *No MHA* feed-forward replacement for the ablation. In the *No MHA* variant, the modality projections, token dimensionality, and mean-pooling operation described in Section IV-C are retained. However, the two multi-head self-attention layers are replaced with two residual feed-forward layers. Each modality token is processed independently before mean pooling, preventing information exchange between modalities. As shown in Table I, removing multi-head attention reduces coverage and increases the collision rate. This indicates that explicitly modeling cross-modal interactions contributes to both inspection performance and navigation robustness.

2) Temporal Memory Ablation: GRU: The baseline policy uses the recurrent processing module described in Section IV-D. To isolate the contribution of temporal memory, the *No GRU* variant replaces the recurrent unit with a feed-forward network of dimensions $512 \to 1536 \to 512$. This replacement contains the same number of trainable parameters as the GRU, ensuring that the comparison isolates recurrence rather than model capacity.

As shown in Table I, removing recurrence produces a slight reduction in coverage but a substantial increase in collision rate. This suggests that temporal memory is particularly important for robust reasoning under partial observability, where individual observations may not fully reveal nearby geometry or the consequences of recent actions.

3) Map Channel Ablation: We evaluate the contribution of the three-channel egocentric map by varying the channels available to the policy while retaining the corresponding map updates and exploration reward signals. Disabled channels are zero-masked, preserving the input dimensions and network architecture across all configurations. We consider the following variants:

- Occupancy only(O): The policy receives only local occupancy information. This results in a modest reduction in coverage and a slight increase in the collision rate relative to SEEIR.
- Occupancy & Visibility Maps (O+V): The policy additionally receives the accumulated surface-visibility channel. However, it doesn't lead to a monotonic improvement; this configuration reduces coverage and increases the collision rate.
- All channels (O+V+Visit): The complete SEEIR policy also receives the visitation channel. This restores performance and outperforms both reduced-map variants in coverage and collision rate.

4) Actuating Pan-Tilt Camera: To evaluate the contribution of active viewpoint control discussed in Section I-A4, we compare SEEIR with a fixed-camera variant. In this variant, the camera joints are held at their neutral positions, and the corresponding actions and joint-state observations are removed from the policy. As shown in Table I, the fixed-camera variant obtains the lowest collision rate but also exhibits the largest reduction in coverage. The lower collision rate may result from the smaller action space, the fixed relative configuration of the depth observations, or less extensive exploration. However, without pan–tilt actuation, the robot must reposition its entire chassis to changes the inspection viewpoint. This limits its ability to observe elevated, lower, or occluded surfaces and demonstrates the importance of active camera control for achieving high inspection coverage.

C. Baseline Comparison

Due to the computational and memory bottleneck of computing point cloud coverage and quality, we save the comparisons for baseline with other algorithms

VIII. Discussion and Future Work

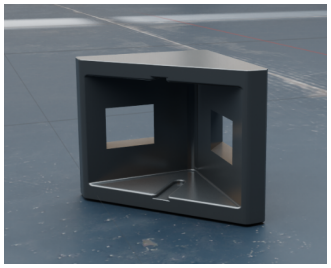
A. Qualitative 3D Reconstruction

While the quantitative evaluation of our method focused on maximizing surface coverage, the ultimate objective of active inspection is to generate high-fidelity digital twins. To validate that our coverage metric translates to downstream reconstruction quality, we qualitatively evaluated the agent's observation history by fusing its depth camera feeds into a global point cloud and comparing it with a point cloud generated from the mesh surface.

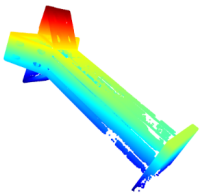
As illustrated in Fig. 4, SEEIR successfully reconstructs complex, non-primitive geometries, such as industrial bracket assemblies. The point cloud generated highlights that the agent effectively navigates to observe the vast majority of the target's surface (green points). The unobserved regions (red points)



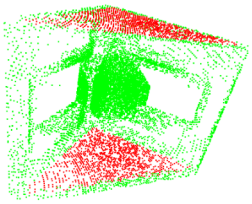
(a) Target: UR10 Mount



(b) Target: Extrusion Bracket



(c) UR10 Mount Coverage



(d) Bracket coverage

Fig. 4: Qualitative evaluation of active inspection. (a, b) industrial mesh targets in Isaac Lab. (c) Point cloud coverage map of the mount; green indicates inspected regions, and red indicates physically occluded surfaces. (d) High-fidelity 3D reconstruction of the bracket generated from the agent's optimized trajectory.

correctly correspond to physically inaccessible areas, such as the bottom faces resting on the floor and internal occlusions.

B. Future Work: 3D Gaussian Splatting and Sim-to-Real

Our immediate future work will focus on evaluating the downstream impact of our coverage-driven trajectories on advanced radiance field reconstructions, specifically 3D Gaussian Splatting (3DGS). We aim to investigate whether our inspection-quality reward naturally correlates with high-fidelity 3DGS outcomes. This evaluation will assess photometric metrics (e.g., PSNR) of rendered scenes. Validating this correlation would confirm that our efficient geometric coverage formulation serves as a robust proxy for radiance field quality, paving the way for targeted photometric reward shaping in future iterations.

Furthermore, transitioning from simulation to real-world deployment introduces significant challenges, including severe state estimation drift, sensor noise, and hardware latency. Future iterations of SEEIR will investigate robust domain randomization techniques and learned state-correction modules to bridge the sim-to-real gap, enabling fully autonomous high-fidelity inspection on physical UGV platforms.

IX. Conclusion

In this work, we presented SEEIR, a novel, semantically-driven deep reinforcement learning framework for autonomous 3D visual inspection. By equipping a mobile ground robot with a simulated Pan-Tilt (PT) camera and formulating a dense, submodular reward structure, our agent learns to simultaneously navigate cluttered environments, circumnavigate complex targets, and actively optimize its camera focal length to maximize high-resolution surface coverage. Through extensive ablation studies, we demonstrated that our multimodal Transformer fusion architecture and our 3-channel egocentric spatial memory are critical for improved convergence. Compared to standard exploration and fixed-trajectory baselines, SEEIR achieves significantly higher inspection coverage on novel, complex geometries. Ultimately, this framework establishes a robust foundation for automated, high-fidelity digital twin generation in unstructured industrial environments.

References

[1] A. Bircher *et al.*, "Structural inspection path planning via iterative viewpoint resampling with application to aerial robotics," in *2015 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2015, pp. 6423–6430.

[2] I. B. d. A. Santos and R. A. Romero, "A deep reinforcement learning approach with visual semantic navigation with memory for mobile robots in indoor home context," *Journal of Intelligent & Robotic Systems*, vol. 104, no. 3, p. 40, 2022.

[3] C. Li *et al.*, "Hrl4in: Hierarchical reinforcement learning for interactive navigation with mobile manipulators," in *Conference on Robot Learning*. PMLR, 2020, pp. 603–616.

[4] A. Faust *et al.*, "Prm-rl: Long-range robotic navigation tasks by combining reinforcement learning and sampling-based planning," in *2018 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2018, pp. 5113–5120.

[5] F. Niroui *et al.*, "Deep reinforcement learning robot for search and rescue applications: Exploration in unknown cluttered environments," *IEEE Robotics and Automation Letters*, vol. 4, no. 2, pp. 610–617, 2019.

[6] Y. Cao *et al.*, "Ariadne: A reinforcement learning approach using attention-based deep networks for exploration," *arXiv preprint arXiv:2301.11575*, 2023.

[7] D. S. Chaplot *et al.*, "Object goal navigation using goal-oriented semantic exploration," *Advances in Neural Information Processing Systems*, vol. 33, pp. 4247–4258, 2020.

[8] ——, "Learning to explore using active neural slam," *arXiv preprint arXiv:2004.05155*, 2020.

[9] G. Malczyk, M. Kulkarni, and K. Alexis, "Semantically-driven deep reinforcement learning for inspection path planning," *IEEE Robotics and Automation Letters*, 2025.

[10] D. Peralta *et al.*, "Next-best view policy for 3d reconstruction," in *European Conference on Computer Vision*. Springer, 2020, pp. 558–573.

[11] B. Mildenhall *et al.*, "Nerf: Representing scenes as neural radiance fields for view synthesis," *Communications of the ACM*, vol. 65, no. 1, pp. 99–106, 2021.

[12] B. Kerbl *et al.*, "3d gaussian splatting for real-time radiance field rendering." *ACM Trans. Graph.*, vol. 42, no. 4, pp. 139–1, 2023.

[13] C. Reiser *et al.*, "Kilonerf: Speeding up neural radiance fields with thousands of tiny mlps," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 14 335–14 345.

[14] Y. Xie *et al.*, "Gauss-mi: Gaussian splatting shannon mutual information for active 3d reconstruction," *arXiv preprint arXiv:2504.21067*, 2025.

[15] M. F. Ginting *et al.*, "SEEK: Semantic reasoning for object goal navigation in real world inspection tasks," *arXiv preprint arXiv:2405.09822*, 2024.

[16] M. Dharmadhikari and K. Alexis, "Semantics-aware predictive inspection path planning," *IEEE Transactions on Field Robotics*, 2025.

[17] C. Wang *et al.*, "Semantic-aware informative path planning for efficient object search using mobile robot," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 51, no. 8, pp. 5230–5243, 2019.

[18] A. Quattrini Li, "Exploration and mapping with groups of robots: Recent trends," *Current Robotics Reports*, vol. 1, no. 4, pp. 227–237, 2020.

[19] T. Dang *et al.*, "Autonomous search for underground mine rescue using aerial robots," in *2020 IEEE Aerospace Conference*. IEEE, 2020, pp. 1–8.

[20] L. Lu *et al.*, "Semantics-aware receding horizon planner for object-centric active mapping," *IEEE Robotics and Automation Letters*, vol. 9, no. 4, pp. 3838–3845, 2024.

[21] K. Nagami *et al.*, "VISTA: Open-vocabulary, task-relevant robot exploration with online semantic gaussian splatting," *IEEE Robotics and Automation Letters*, 2026.

[22] P. Chen *et al.*, "Learning active camera for multi-object navigation," *Advances in Neural Information Processing Systems*, vol. 35, pp. 28 670–28 682, 2022.
[23] G. Malczyk, M. Kulkarni, and K. Alexis, "Reinforcement learning for active perception in autonomous navigation," *arXiv preprint arXiv:2602.01266*, 2026.
[24] S. S. Sandha *et al.*, "Eagle: End-to-end deep reinforcement learning based autonomous control of ptz cameras," in *Proceedings of the 8th ACM/IEEE Conference on Internet of Things Design and Implementation*, 2023, pp. 144–157.
[25] Z. Yang *et al.*, "Active visual perception enhancement method based on deep reinforcement learning," *Electronics*, vol. 13, no. 9, p. 1654, 2024.
[26] ——, "Multi-agent hierarchical reinforcement learning for ptz camera control and visual enhancement," *Electronics*, vol. 14, no. 19, p. 3825, 2025.
[27] F. Lu *et al.*, "Active tracking of moving objects with speed variations using a novel ptz camera-based machine vision technique," *Elektronika ir Elektrotechnika*, vol. 31, no. 2, pp. 60–70, 2025.
[28] W. Wu *et al.*, "Spatial resolution ensured configuration planning for visual inspection using pan-tilt-zoom camera-equipped robots," *IEEE Access*, 2025.
[29] Y. Tan *et al.*, "An adaptive crack inspection method for building surface based on bim, uav and edge computing," *Automation in Construction*, vol. 157, p. 105161, 2024.
[30] I. Jovančević *et al.*, "Automated exterior inspection of an aircraft with a pan-tilt-zoom camera mounted on a mobile robot," *Journal of Electronic Imaging*, vol. 24, no. 6, pp. 061 110–061 110, 2015.
[31] S. Hughes *et al.*, "Multi-robot planning for filming groups of moving actors leveraging submodularity and pixel density," in *2024 IEEE 63rd Conference on Decision and Control (CDC)*. IEEE, 2024, pp. 5401–5408.
[32] R. Bachmann *et al.*, "MultiMAE: Multi-modal multi-task masked autoencoders," 2022.
[33] O. M. Team *et al.*, "Octo: An open-source generalist robot policy," *arXiv preprint arXiv:2405.12213*, 2024.
[34] A. Serrano-Muñoz *et al.*, "skrl: Modular and flexible library for reinforcement learning," *Journal of Machine Learning Research*, vol. 24, no. 254, pp. 1–9, 2023.
[35] L. Espeholt *et al.*, "IMPALA: Scalable distributed Deep-RL with importance weighted actor-learner architectures," in *International conference on machine learning*. PMLR, 2018, pp. 1407–1416.